

Foundation (Jalapeno)

- Q1.** A function $f(x)$ has a vertical shift of 2 units down. What is the new function?
(A) $f(x) - 2$
(B) $f(x) + 2$
(C) $f(x - 2)$
(D) $f(x + 2)$
(E) $2f(x)$
- Q2.** If $g(x) = x^2$, what is the equation of $g(x)$ after a horizontal shift of 3 units to the right?
(A) $g(x - 3)$
(B) $g(x + 3)$
(C) $g(x)^2 + 3$
(D) $g(x)^2 - 3$
(E) $g(3x)$
- Q3.** What is the effect of reflecting $f(x) = x^2$ over the y-axis?
(A) $f(x)$ becomes $f(-x)$
(B) $f(x)$ becomes $-f(x)$
(C) $f(x)$ becomes $f(x) + 1$
(D) $f(x)$ becomes $f(x) - 1$
(E) $f(x)$ remains the same
- Q4.** If a function $h(x)$ has a stretch factor of 2, what is the new function?
(A) $2h(x)$
(B) $h(2x)$
(C) $h(x/2)$
(D) $h(x) + 2$
(E) $h(x) - 2$
- Q5.** A scientist is studying the trajectory of a planet. The original function is $d(t) = t^2$. What is the equation after a reflection over the x-axis?
(A) $-d(t)$
(B) $d(t) + 1$
(C) $d(t) - 1$
(D) $d(-t)$
(E) $-d(-t)$
- Q6.** If a function $f(x) = x^2$ has a compression factor of $1/2$, what is the new function?
(A) $f(2x)$
(B) $f(x/2)$
(C) $f(x) + 1/2$
(D) $f(x) - 1/2$
(E) $2f(x)$
- Q7.** What is the effect of shifting $f(x) = x^2$ 1 unit up and 2 units to the left?
(A) $f(x + 2) + 1$
(B) $f(x - 2) + 1$
(C) $f(x + 2) - 1$
(D) $f(x - 2) - 1$
(E) $f(x) + 2$
- Q8.** If $g(x) = |x|$, what is the equation of $g(x)$ after a reflection over the y-axis and a shift of 1 unit down?
(A) $-g(x) - 1$
(B) $-g(-x) - 1$
(C) $g(x) - 1$
(D) $g(-x) - 1$
(E) $g(x) + 1$
- Q5.** A scientist is studying the trajectory of a planet. The orig-

Open Ended Questions (FRQ)

- Q9.** Describe the transformation of $f(x) = x^2$ into $g(x) = -2f(x - 1) + 3$. Use [answer] for each part. What are the vertical and horizontal shifts, stretch/compression factor, and reflection? Show all work and explain your reasoning clearly.
- Q10.** Compare the graphs of $f(x) = x^2$ and $g(x) = f(2x) + 1$. How do the graphs differ in terms of shifts, stretches, and reflections? Provide a sketch of both graphs with [answer] labels and explain the transformation used to obtain $g(x)$ from $f(x)$. Use [answer] for each part.